7.5 Shock absorber

The spring system prevents the travelers and truckloads that do not cause any shocks when climbing up the sudden disturbance on the road. However, if the spring are stiffer than expected, it will not absorb the road shock. The spring will keep on oscillate even after crossing the disturbance till the energy is completely die out. Hence, to quickly dissipate the stored energy in springs, dampers are used.

According to the working principle shock absorber are classified into three types they are,

1. Friction type
2. Hydraulic type
3. Air type

7.5.1 Friction type

When one body is allowed to slide over the other, the surface of one body offers some resistance to the movement of the other body on it. This resisting force is called friction force. In leaf spring, the interleaf friction is act as a damper. It is not widely used with other types of springs.

7.5.2 Hydraulic type

This type of damper is most widely used in the suspension system. It works based on Pascal'slaw. They are further classified into two types.

(i) Pistontype
(ii) Telescopic type

Telescopic type

It works on the basic principle of Pascal's law. Damping is obtained as high-velocity fluid passing through a tiny hole. This system is also called telescopic shock absorber and because of use of hydraulic, it is also called as hydraulic damper.

Construction

Figure 7.5.2.2 shows the construction details of this shock absorber. The head is fixed inside the cylinder. The piston and two way valve located on the top portion of the Cylinder is assembled with piston rod for up and down motion. The other end of the piston rod has eye and it is attached to the chassis frame. On the bottom portion of the cylinder another two way valve named as foot valve is attached. In this two way valve, one valve is used for fluid inlet and other valve is used for fluid outlet. Cylinder is filled with fluid.

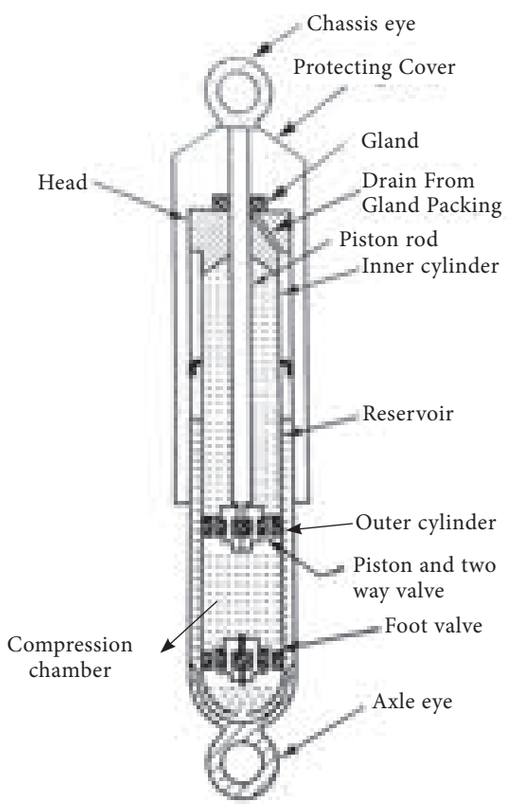


Figure 7.5.2.2 Telescopic type shock absorber

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